

1. Part 1: Device Sizing Using SA

Design Specifications

We are required to design a CD amplifier that has ideal current source load with the parameters below

Parameter	Value
Input transistor	PMOS
L	$1\mu m$
V^*	200 mV
Quiescent (DC) input voltage	0 V
Supply	1.8 V
Current consumption	$10\mu A$

A) Sizing Assistant Extraction

Import: Plot 1B OK ?

LUT Settings

LUT pmos_03v3 ?
Corner TT All ?
Temp (°C) 27.0 All ?
Frequency 1 ?
ID 10u ?
gm/ID 10 ?
L 1u ?
VDS VGS ?
VSB 0 ?
Stack 1 ?

Results:

Name		TT-27.0
4	W	19.36u

Figure 1: ADT sizing

2. Part 2: CD Amplifier

2.1 1. OP (Operating Point) Analysis

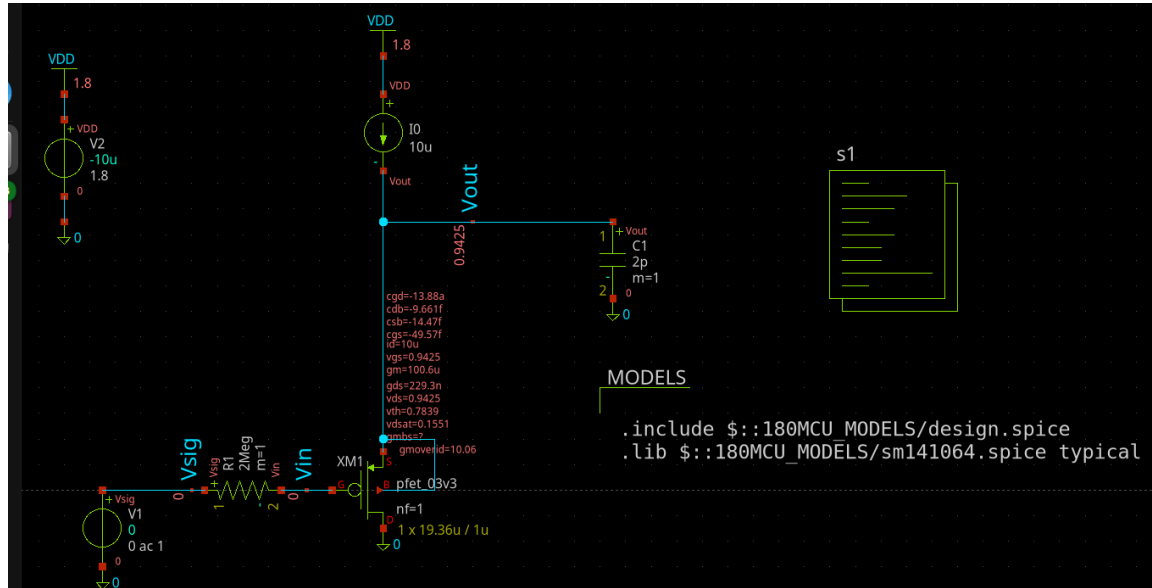


Figure 2

Saturation Check

Q: Check that the transistor operates in saturation.

Yes, $V_{DS} > V_{DSSAT}$

2.2 2. AC Analysis

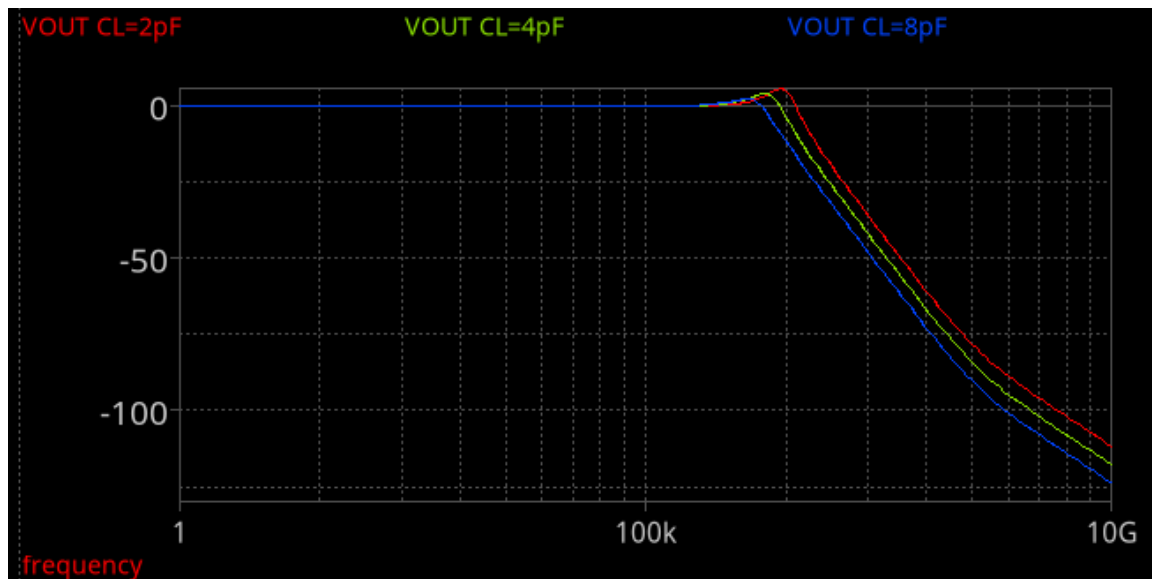


Figure 3: Frequency Domain Peaking with variable C_L

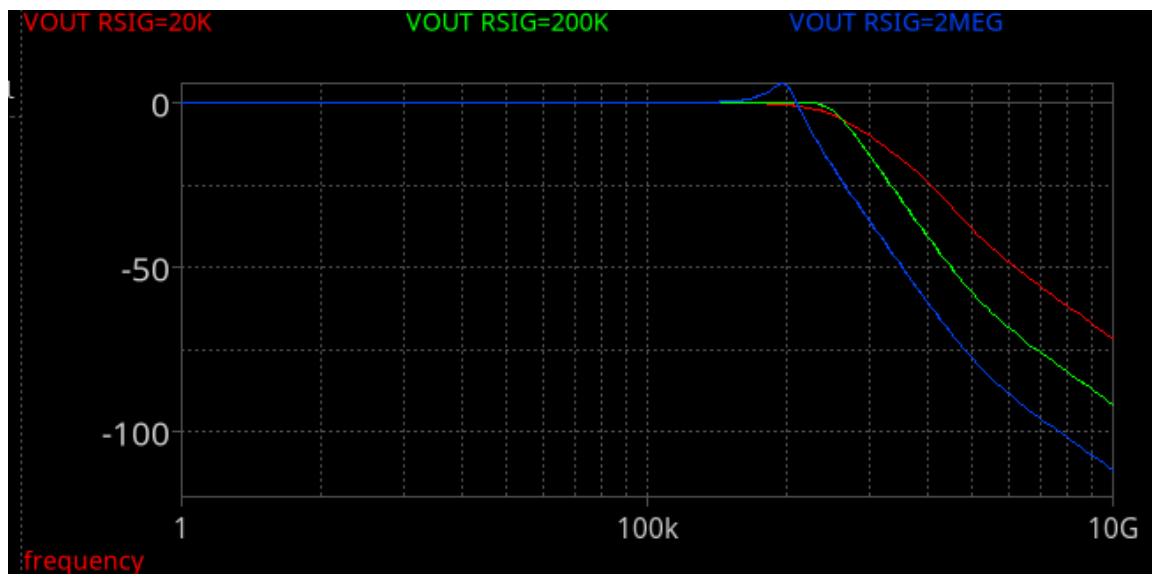


Figure 4: Frequency Domain Peaking with variable R_{sig}

Peaking and Quality Factor

Q: Do you notice frequency domain peaking? Analytically calculate the quality factor. Is the system underdamped or overdamped?

Yes, There is peaking. This is due to the complex conjugate and high frequency nature of the Common Drain's poles. Furthermore, They are very near, making

Miller theorem and OCTC method invalid.

$$\omega_{p,in} = \frac{1}{(C_{gd} + C_{gs}(1 - A_v))R_{sig}}$$

$$\omega_{p,out} = \frac{1}{(C_L + C_{sb} + C_{gs}(1 - 1/A_v))(\frac{1}{g_m})}$$

The two poles are close and usually complex

$$\frac{V_{out}}{V_{sig}} = \frac{1 + s/w_z}{(1 + \frac{s}{w_{p1}})(1 + \frac{s}{w_{p2}})} = \frac{1 + s/w_z}{1 + b_1s + b_2s^2}$$

We usually represent a second order system in this form:

$$\frac{V_{out}}{V_{sig}} = \frac{1 + s/w_z}{s^2 + \frac{w_o}{Q}s + w_o^2}$$

By neglecting R_s and CLM we get:

$$b_1 = C_{gd}R_{sig} + \frac{C_{gs} + C_L}{g_m}$$

$$b_2 = \frac{((C_{gs} + C_{gd})C_L + C_{gs}C_{gd})}{g_m}R_{sig}$$

$$w_z = \frac{g_m}{C_{gs}} \quad w_o = 1/\sqrt{b_2} \quad Q = \frac{\sqrt{b_2}}{b_1}$$

Getting Q

Plugging in the values from the OP:

$$b_1 \approx 20.40 \text{ ns}$$

$$b_2 \approx 1.972 \times 10^{-15} \text{ s}^2$$

$$\omega_o = \frac{1}{\sqrt{1.972 \times 10^{-15}}} \approx 22.52 \text{ Mrad/s}$$

$$Q = \frac{\sqrt{1.972 \times 10^{-15}}}{20.40 \times 10^{-9}} = 2.18 > 0.707$$

Meaning that the system is underdamped ($\zeta < 0.5$) the poles form a complex conjugate pair. We find the true poles by calculating the roots of the characteristic equation $b_2s^2 + b_1s + 1 = 0$:

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CD Amplifier AC Sim Results
CL = 2p, Gain Peak (dB) = 5.9966
CL = 4p, Gain Peak (dB) = 4.33344
CL = 8p, Gain Peak (dB) = 2.3851
RES = 20k, Gain Peak (dB) = -0.0197724
RES = 200k, Gain Peak (dB) = 0.121317
RES = 2meg, Gain Peak (dB) = 5.9966

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Figure 5: Peaking Results

Peaking vs. C_L

Q: Perform parametric sweep: $C_L = 2p, 4p, 8p$. Report the peaking vs C_L and comment.

From the expression above, We find that:

$$Q \propto \frac{1}{\sqrt{C_L}}$$

Increasing C_L means decreasing Q leading to less peaking and ringing as shown in Figure 3, at the cost of lower bandwidth.

Peaking vs. R_{sig}

Q: Perform parametric sweep: $R_{sig} = 20k, 200k, 2M$. Report the peaking vs R_{sig} and comment.

Increasing R_{sig} means increasing Q , leading to more peaking and ringing. This happens because increasing the source resistance pushes the input pole to a lower frequency, bringing it closer to the output pole and creating complex conjugate roots. Here is the mathematical justification based on the formulas:

- In the b_1 equation, the term $C_{gd}R_{sig}$ is practically negligible because C_{gd} is extremely small (in the attofarad range). Thus, the denominator remains roughly constant:

$$b_1 \approx \frac{C_{gs} + C_L}{g_m}$$

- In the b_2 equation, the term is directly proportional to R_{sig} :

$$b_2 \propto R_{sig}$$

- Since $Q = \frac{\sqrt{b_2}}{b_1}$, dividing the numerator by the roughly constant denominator yields a direct proportionality to the square root of the source resistance:

$$Q \propto \sqrt{R_{sig}}$$

2.3 3. Transient Analysis

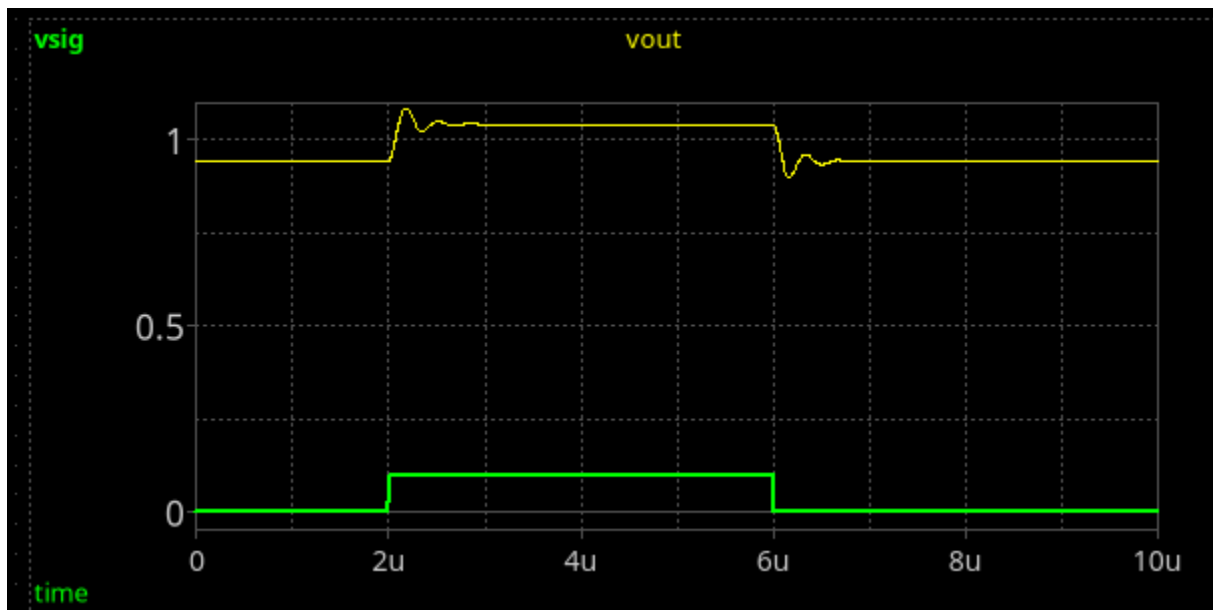


Figure 6: Vsig and Vout Overlaid

DC Shift and Ringing

Q: Calculate the DC voltage difference (DC shift) between V_{in} and V_{out} . What is the relation to V_{GS} ? How to shift the signal down? Do you notice time domain ringing?

The DC shift between V_{in} and V_{out} is nearly equal V_{SG} even though both signals suffer from ringing. To shift the DC level down instead of up, use an NMOS transistor where $V_{out} = V_{in} - V_{GS}$.

Yes, Ringing occurs before peaking. So, if V_{out} suffers from peaking, it definitely suffers from ringing.



Figure 7: Parametric Sweep (CL): Vout vs. Time

CD Amplifier Transient Sim Results

CL = 2p, Overshoot (%) = 43.0232

CL = 4p, Overshoot (%) = 34.1077

CL = 8p, Overshoot (%) = 22.4211

Rsig = 20k, Overshoot (%) = 0

Rsig = 200k, Overshoot (%) = 6.58342

Rsig = 2meg, Overshoot (%) = 43.0232

Figure 8: Overshoot Results

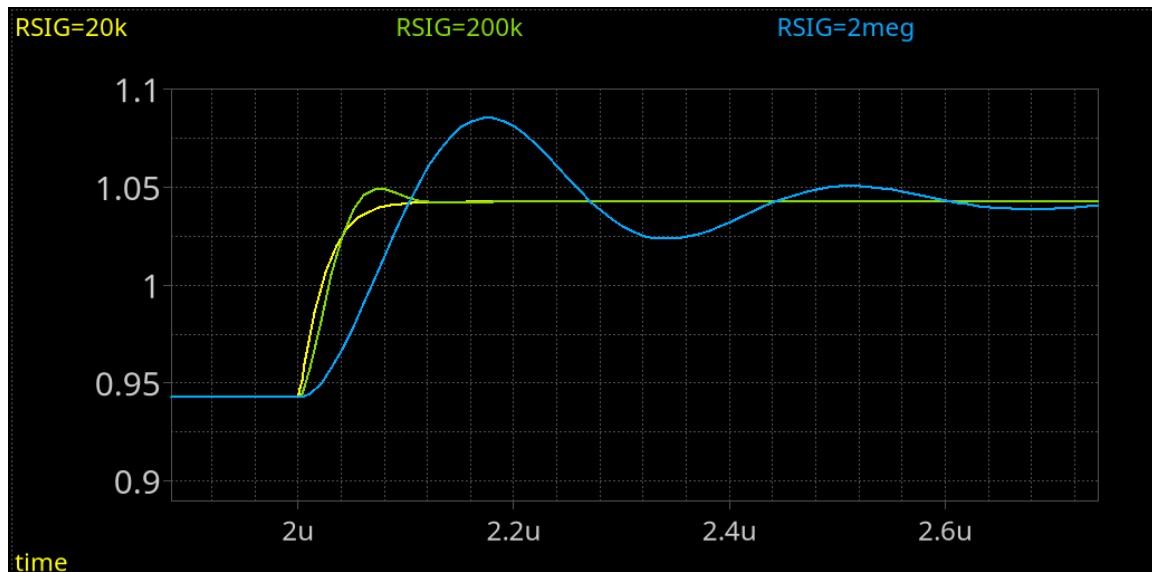
Overshoot vs. C_L

Q: Perform parametric sweep: $C_L = 2p, 4p, 8p$. Report the overshoot vs C_L and comment.

$$\text{Overshoot (\%)} = \frac{V_{peak} - V_{final}}{V_{final} - V_{initial}} \times 100$$

$$\text{Overshoot (\%)} = 100e^{-\frac{\pi}{\sqrt{4Q^2-1}}}$$

As shown, As C_L increases, leading to less Q , which means less overshoot.

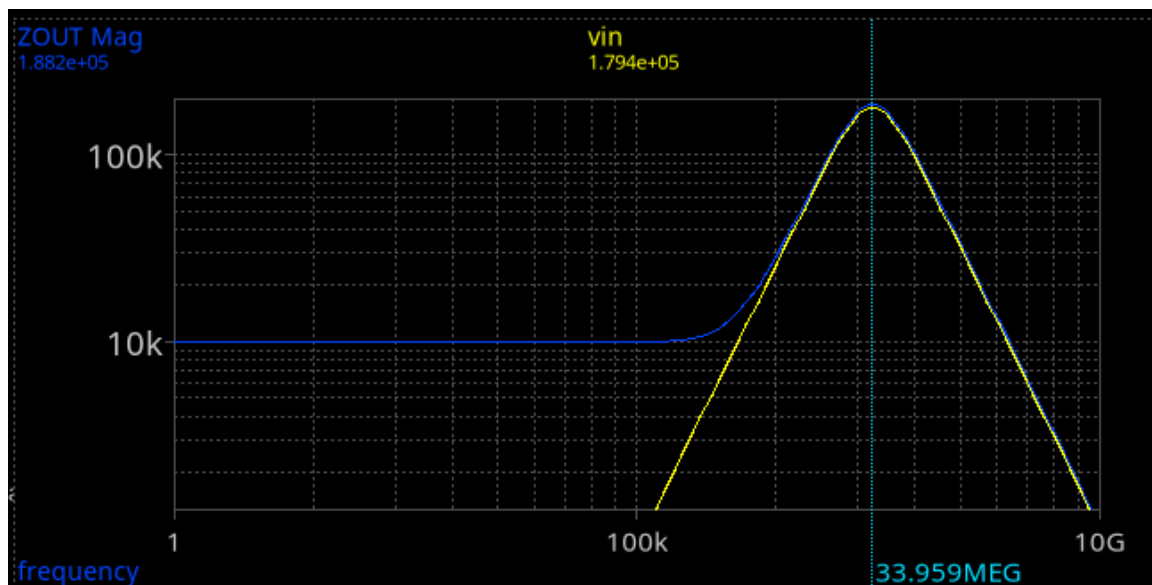
Figure 9: Parametric Sweep (R_{sig}): V_{out} vs. Time

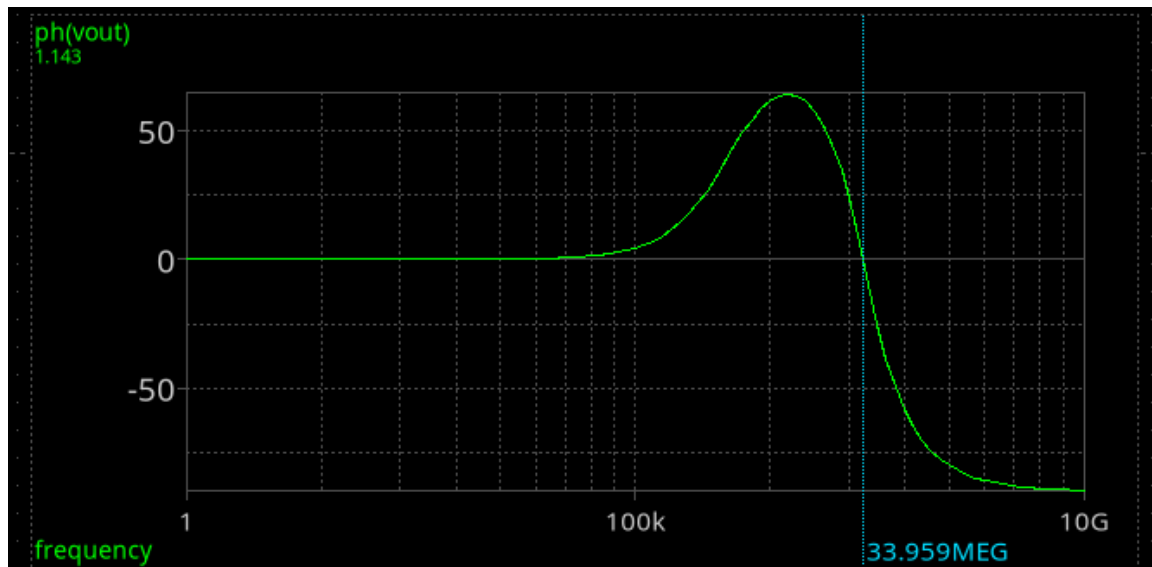
Overshoot vs. R_{sig}

Q: Perform parametric sweep: $R_{sig} = 20k, 200k, 2M$. Report the overshoot vs R_{sig} and comment.

As R_{sig} increases, Q rises leading to more overshoot.

2.4 4. Z_{out} (Inductive Rise)

Figure 10: $|Z_{out}|$ vs frequency

Figure 11: $\angle Z_{out}$ vs frequency

Inductive Rise Analysis

Q: Do you notice an inductive rise? Why? Does Z_{out} fall at high frequency? Why?

Yes, Inductive rise means the output impedance increases with frequency. At low frequencies, R_{out} sees $\frac{1}{g_m}$. As frequency rises, $\frac{1}{g_m}$ is shorted with C_{gs} , seeing R_{sig} , which is usually greater than $\frac{1}{g_m}$. However, at very high frequencies, C_{gd} begins to short R_{sig} , preventing Z_{out} from reaching both the casual phase of $+90$ for inductance and R_{sig} of $2M$.

CD Amplifier Z_{out} Sim Results

$Z_{out} @ 1kHz = 9917.68$

$Z_{out} \text{ peak (max)} = 188181$

Figure 12: Z_{out} results

Q: Analytically calculate the zeros, poles, and magnitude at low/high frequency for Z_{out} . Compare the results in a table.

$$Z_{out} = \frac{v_x}{i_x} = \frac{1}{g_m} \left(\frac{1 + sR_{sig}C_{gs}}{1 + s\frac{C_{gs}}{g_m}} \right)$$

$$R_{sig} = 2 \text{ M}\Omega$$

$$C_{gs} \approx 49.57 \text{ fF}$$

$$g_m \approx 100.8 \text{ }\mu\text{S}$$

1. Zero Calculation (f_z)

$$f_z = \frac{1}{2\pi R_{sig} C_{gs}}$$

$$f_z = \frac{1}{2\pi (2 \times 10^6) (49.57 \times 10^{-15})}$$

$$f_z = \frac{1}{622.91 \times 10^{-9}}$$

$$f_z \approx 1.605 \text{ MHz}$$

2. Pole Calculation (f_p)

$$f_p = \frac{g_m}{2\pi C_{gs}}$$

$$f_p = \frac{100.8 \times 10^{-6}}{2\pi (49.57 \times 10^{-15})}$$

$$f_p = \frac{100.8 \times 10^{-6}}{311.46 \times 10^{-15}}$$

$$f_p \approx 323.6 \text{ MHz}$$

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No. of Data Rows : 1001
binary raw file "cd_amp_zout.raw"
zout_low_freq      = 9.917681e+03
zout_high_freq     = 1.901548e+05 at= 3.548134e+07
sim_zero           = 1.179268e+06
```

Figure 13: Simulated Zero and Zout at low and high frequency

Parameter	Hand Calculation (Theory)	Ngspice Simulation	Deviation / Notes
Z_{out} (Low Freq)	$1/g_m \approx 9.92 \text{ k}\Omega$	9.917 k Ω	< 0.1% error
Z_{out} (High Freq)	$R_{sig} = 2 \text{ M}\Omega$	188.18 k Ω (Peak)	Masked by Resonance
f_z (Zero Frequency)	$\frac{1}{2\pi R_{sig} C_{gs}} \approx 1.605 \text{ MHz}$	1.179 MHz	$\approx 26.5\%$ error

Table 1: Comparison of Theoretical vs. Simulated AC Parameters

2.5 5. Bonus: Solving the Peaking/Ringing Problem

Compensation Network

Q: How to solve the peaking/ringing problem? (Read Section 4.4 and try to implement the compensation network)

The trivial solutions is to either increase C_L or decrease R_{sig} . However, Increasing the load capacitance reduces the bandwidth (which is against the goal I implemented the CD for, which is increasing the bandwidth by providing low impedance node). Furthermore, R_{sig} is the output impedance of the previous amplifier in the chain. Decreasing Rout reduces the gain of the previous stage (which should be a black box for me when designing the CD)

The trick here is to implement a compensation network to further away the two poles in my system and make them real on the σ axis without changing BW or decreasing Rsig.

Its idea is to cancel the negative admittance (the negative resistance and capacitance, which generate energy rather than dissipating/storing it) by adding equivalent positive series RC in parallel.

First, find a common denominator for the series impedance Z_{in} :

$$Z_{in} = \frac{s(C_{gs} + C_L) + g_m}{s^2 C_{gs} C_L}$$

Invert it to find the parallel admittance Y_{in} :

$$Y_{in} = \frac{1}{Z_{in}} = \frac{s^2 C_{gs} C_L}{s(C_{gs} + C_L) + g_m}$$

Pull out a series equivalent capacitor $C_2 = \frac{C_{gs} C_L}{C_{gs} + C_L}$ to split the fraction:

$$Y_{in} = sC_2 \left(1 - \frac{g_m}{s(C_{gs} + C_L) + g_m} \right)$$

Multiply the sC_2 through:

$$Y_{in} = sC_2 + \frac{-sC_2 g_m}{s(C_{gs} + C_L) + g_m}$$

Finally, divide the numerator and denominator of the second term by $-sC_2 g_m$ to

match the standard admittance form $Y = \frac{1}{R + \frac{1}{sC}}$:

$$Y_{in} = sC_2 + \frac{1}{-\left(\frac{C_{gs} + C_L}{C_2 g_m}\right) - \frac{1}{sC_2}}$$

$$C_1 \approx 48.37 \text{ fF}$$

$$R_1 \approx 421.2 \text{ k}\Omega$$

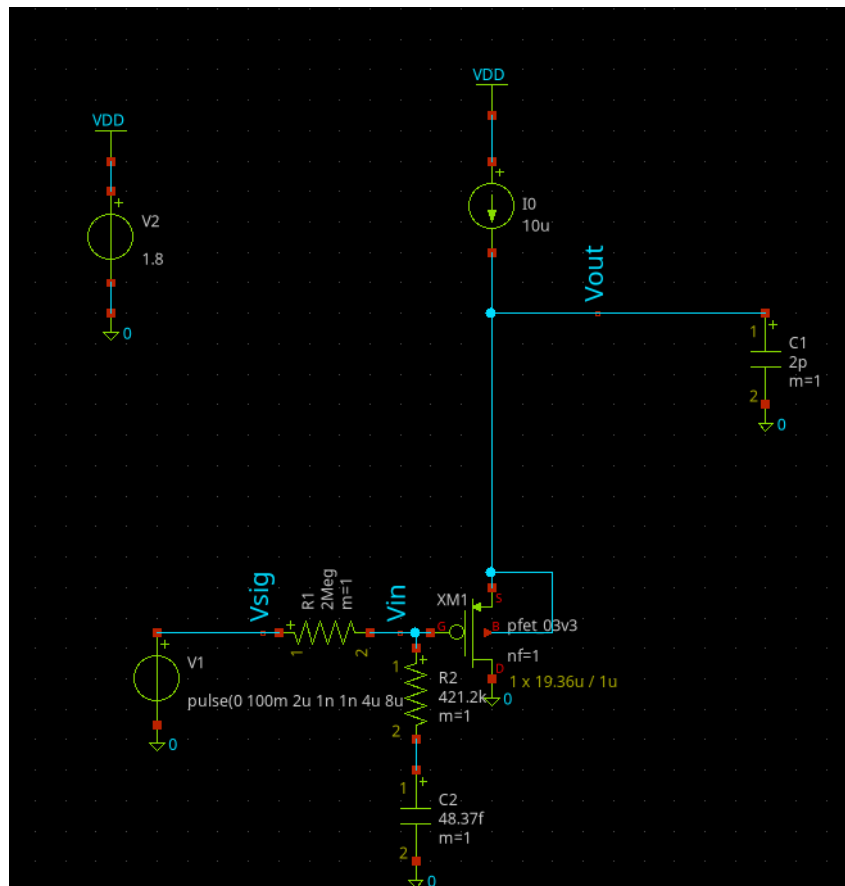


Figure 14: Schematic after adding the compensation network

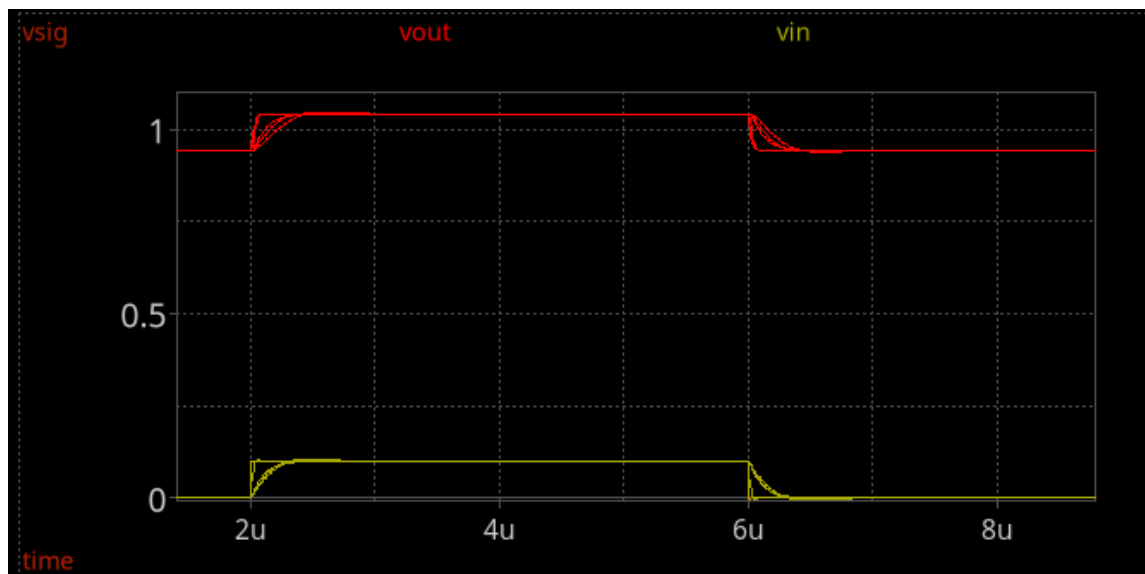


Figure 15: Vout and Vsig overlaid

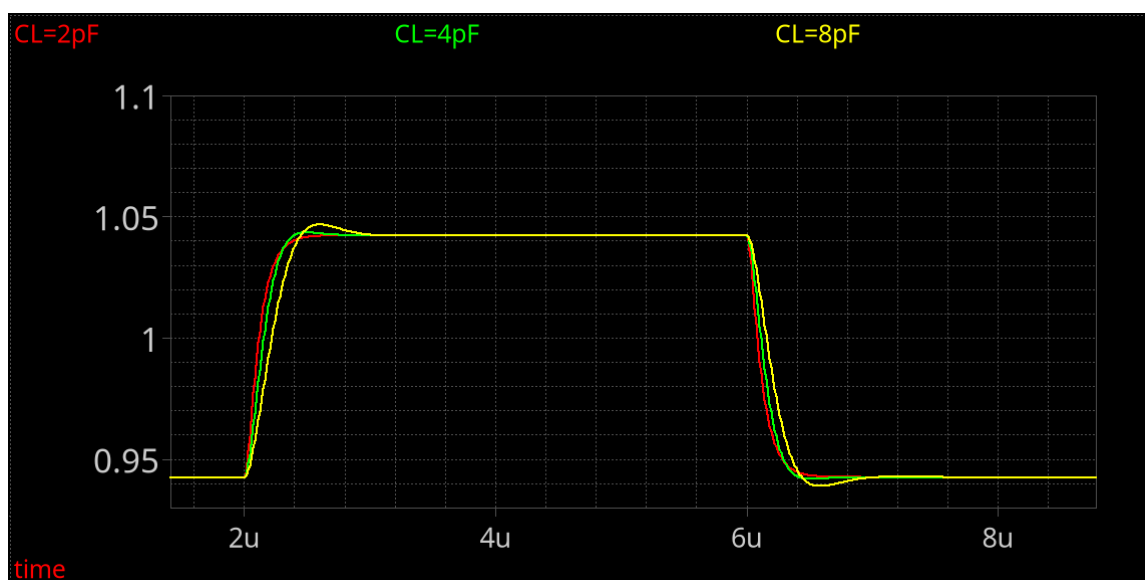


Figure 16: Vout with parametric CL

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CD Amplifier Transient Sim Results
CL = 2p, Overshoot (%) = 0
CL = 4p, Overshoot (%) = 1.16956
CL = 8p, Overshoot (%) = 4.46879
Rsig = 20k, Overshoot (%) = 0
Rsig = 200k, Overshoot (%) = 0.280615
Rsig = 2meg, Overshoot (%) = 0
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Figure 17: Overshoot results

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CD Amplifier AC Sim Results
CL = 2p, Gain Peak (dB) = -0.0197724
CL = 4p, Gain Peak (dB) = -0.0197724
CL = 8p, Gain Peak (dB) = -0.0197724
RES = 20k, Gain Peak (dB) = -0.0197724
RES = 200k, Gain Peak (dB) = -0.0197724
RES = 2meg, Gain Peak (dB) = -0.0197724
```

Figure 18: Peaking results

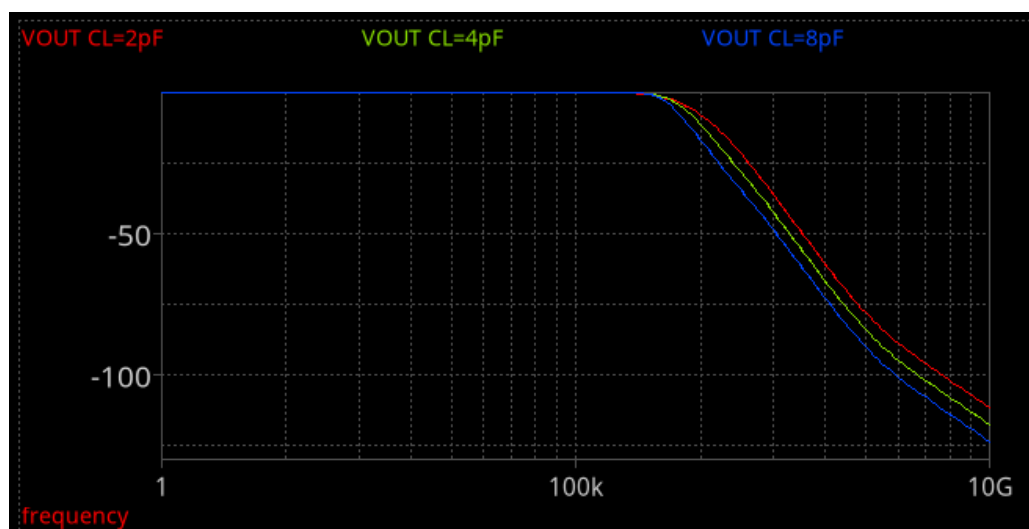


Figure 19: Frequency response